

Data ethics and privacy

INTRODUCTION TO DATA



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Basics of data ethics

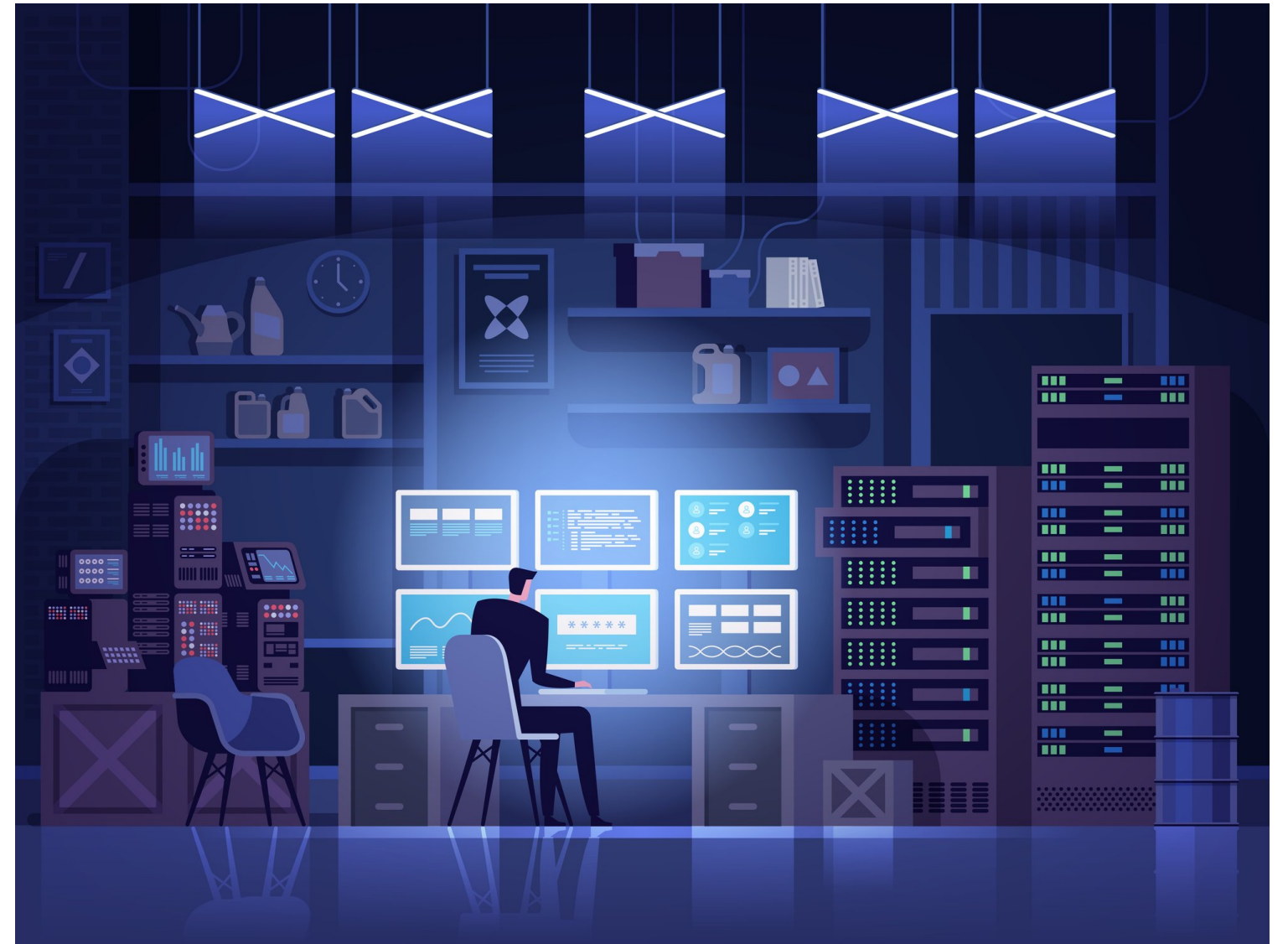
- The advantages of data come with responsibilities
- Data ethics ensures that data is collected, stored, and used ethically



Principles of data ethics

Five principles¹

1. Permission for data collection
2. Transparency about the plan
3. Privacy of data
4. Good intentions
5. Consider the **outcome**



¹ Harvard Business School, 2021

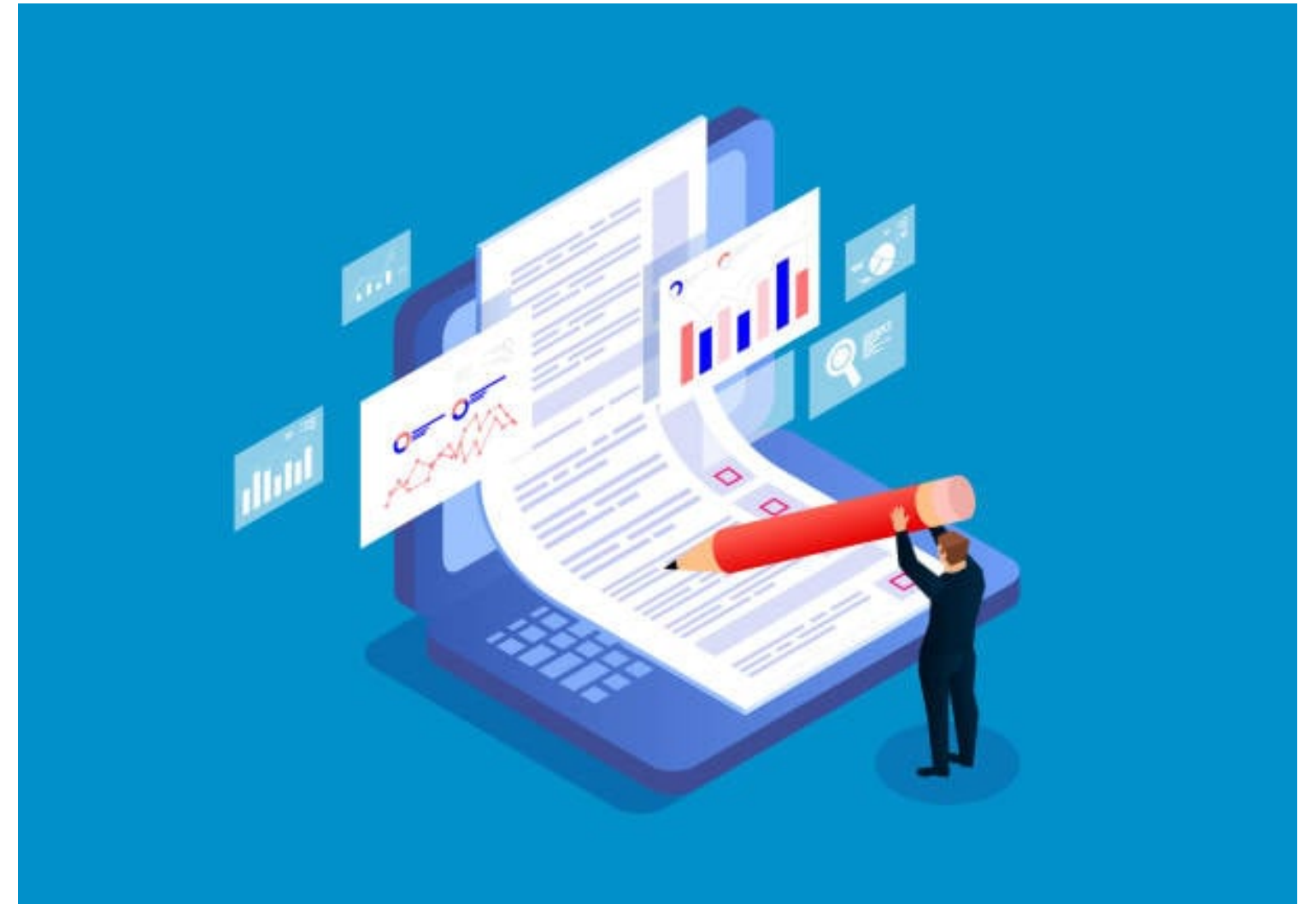
Permission and transparency

Permission

- Ask before collecting
- Users are in control of their data

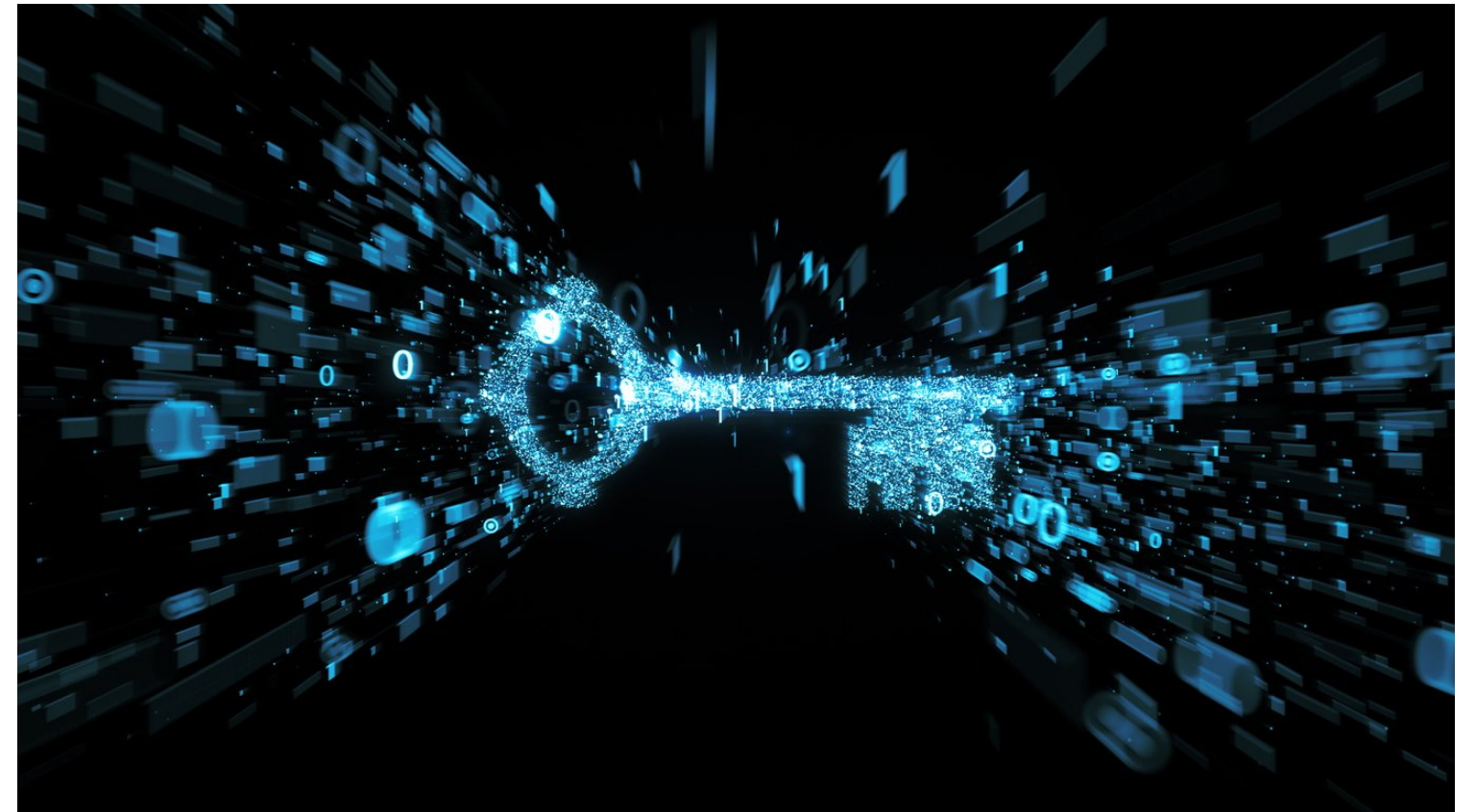
Transparency

- Be transparent of how you plan to use, store, and collect data
- Lack of transparency may lead to reputation and legal damage



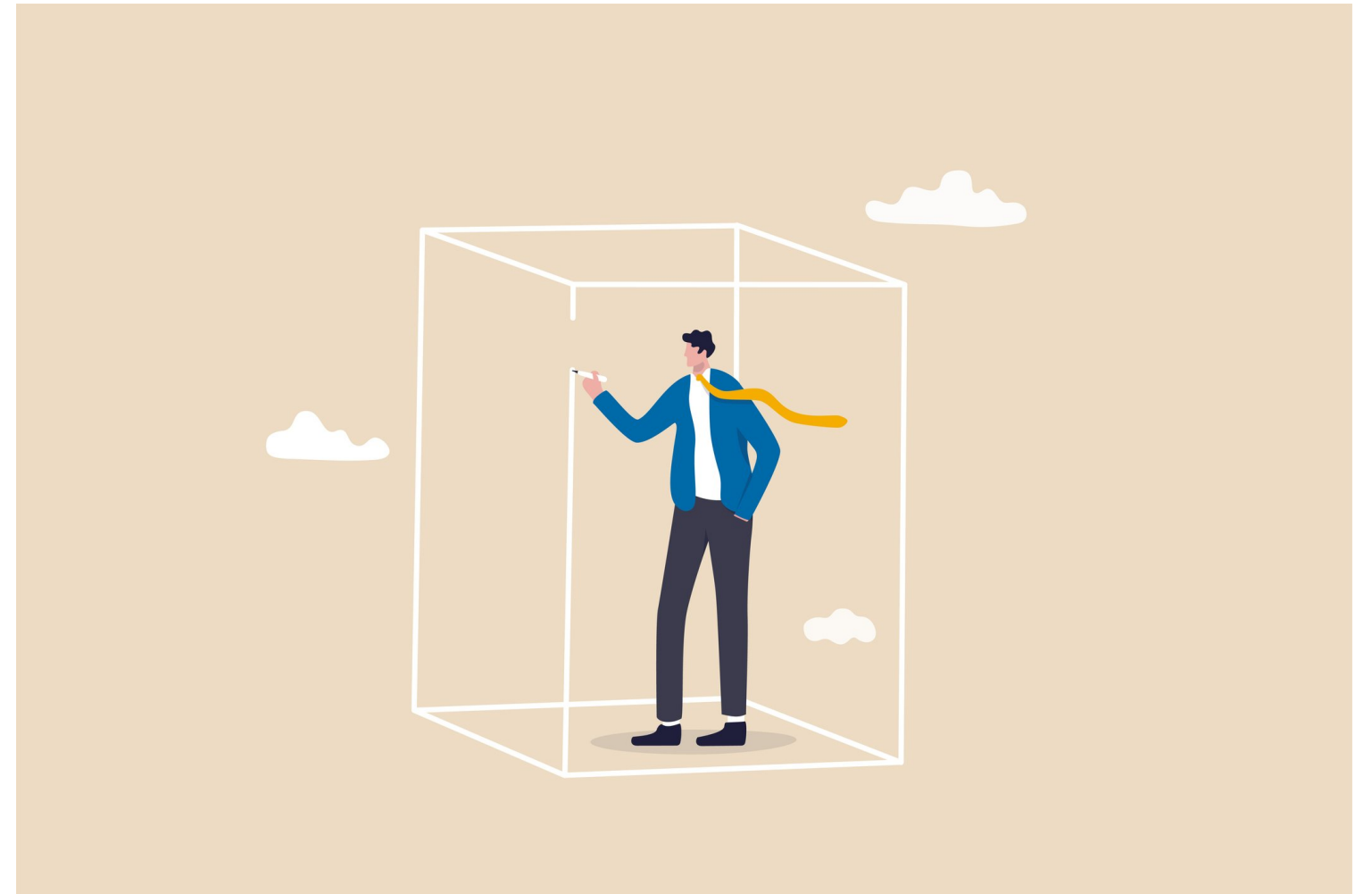
Data privacy

- Privacy refers to secluding (information about) yourself
- Personal identifiable information (PII) must remain confidential
 - Full name
 - Date of birth
 - Bank account
 - ID card/Passport number
 - Phone number
 - ...



Privacy protection

- Strong passwords
- Up-to-date operating systems
- Cautionary Internet browsing
- **Individual responsibility**
- Prevention by:
 - Limit sharing sensitive data
 - (Pseudo-)data anonymization



Intentions and outcome



Intentions

- Data is collected for the right reasons
- Question yourself about the reasons you collect data

Outcome

- Are there consequences of my actions?
- Protecting vulnerable populations

Let's practice!

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Data life cycle

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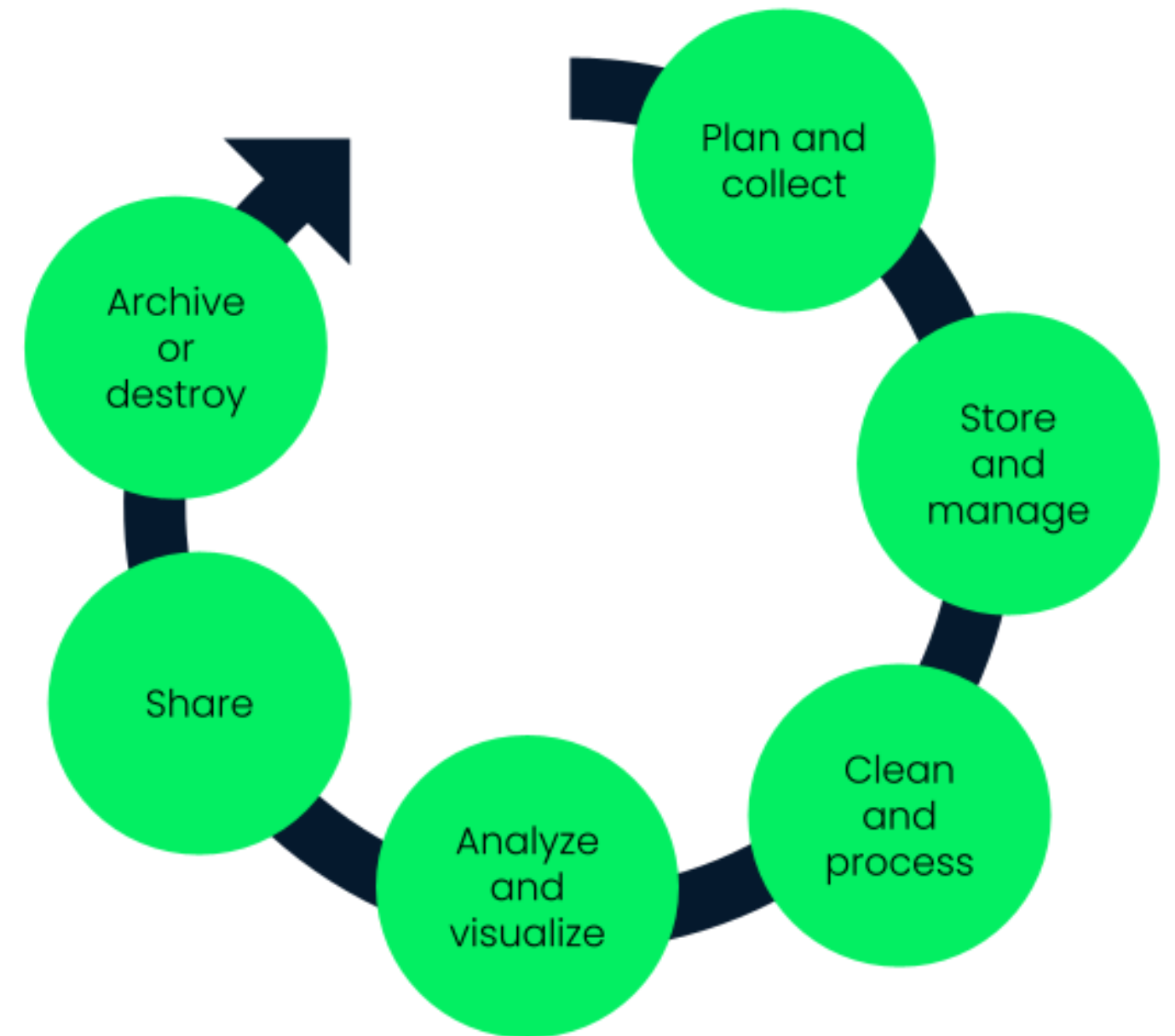


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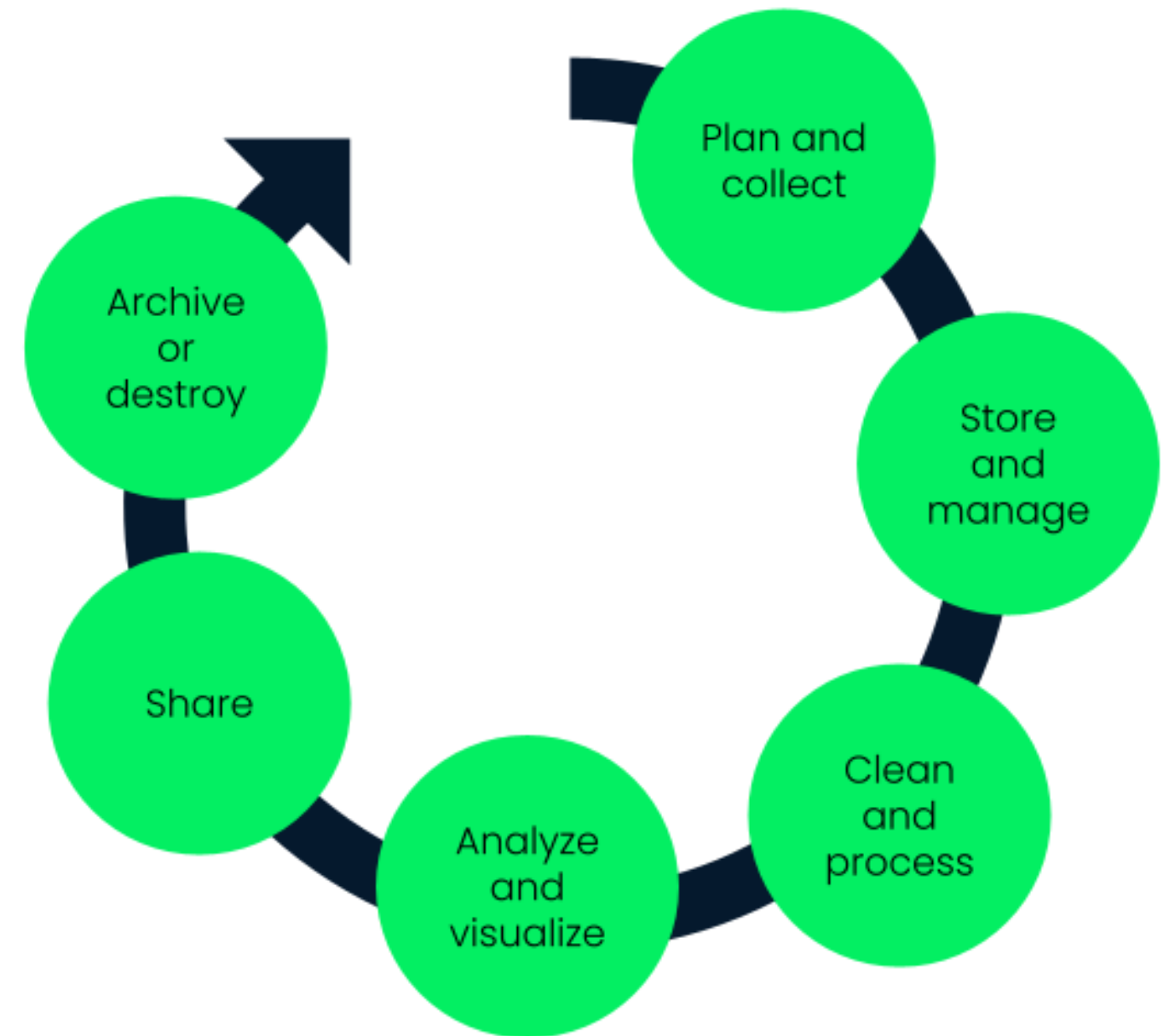
What is the data life cycle?

- Planning and collecting
- Storing and managing
- Cleaning and processing
- Analyzing and visualizing
- Sharing
- Archiving/destroying

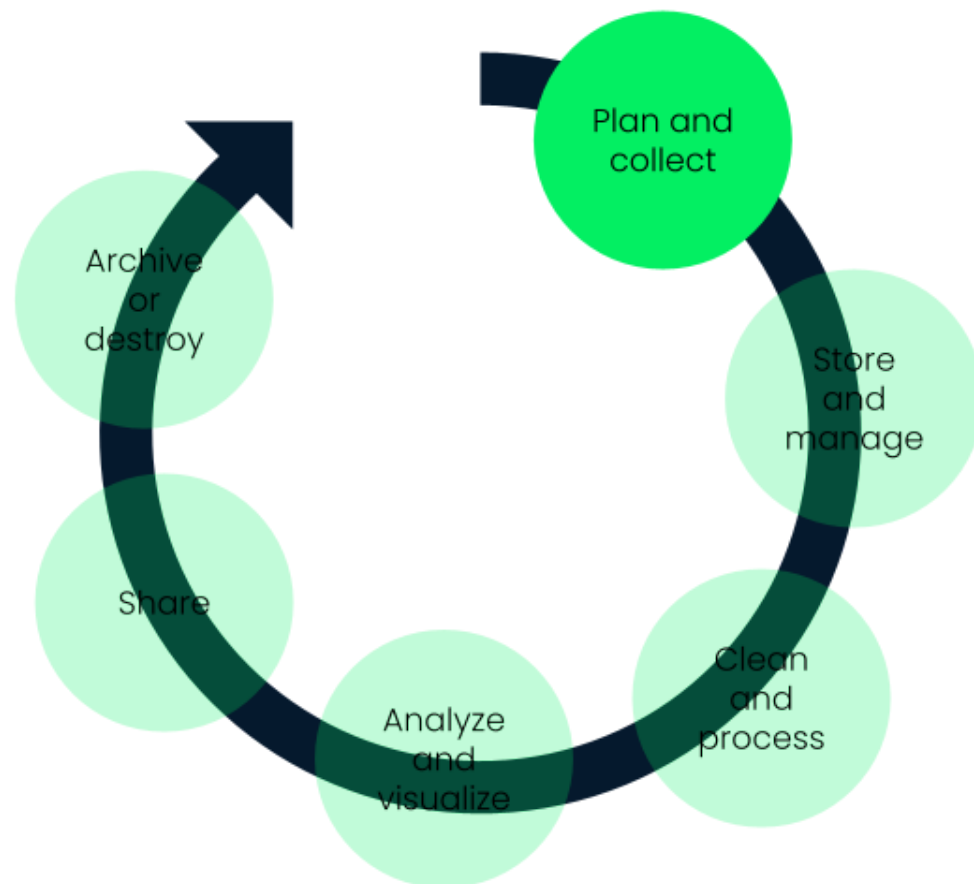


Why is the data life cycle important?

- Ensure data is regulated responsibly
- Identify potential areas for improvement
- Improve efficiency and effectiveness of operations

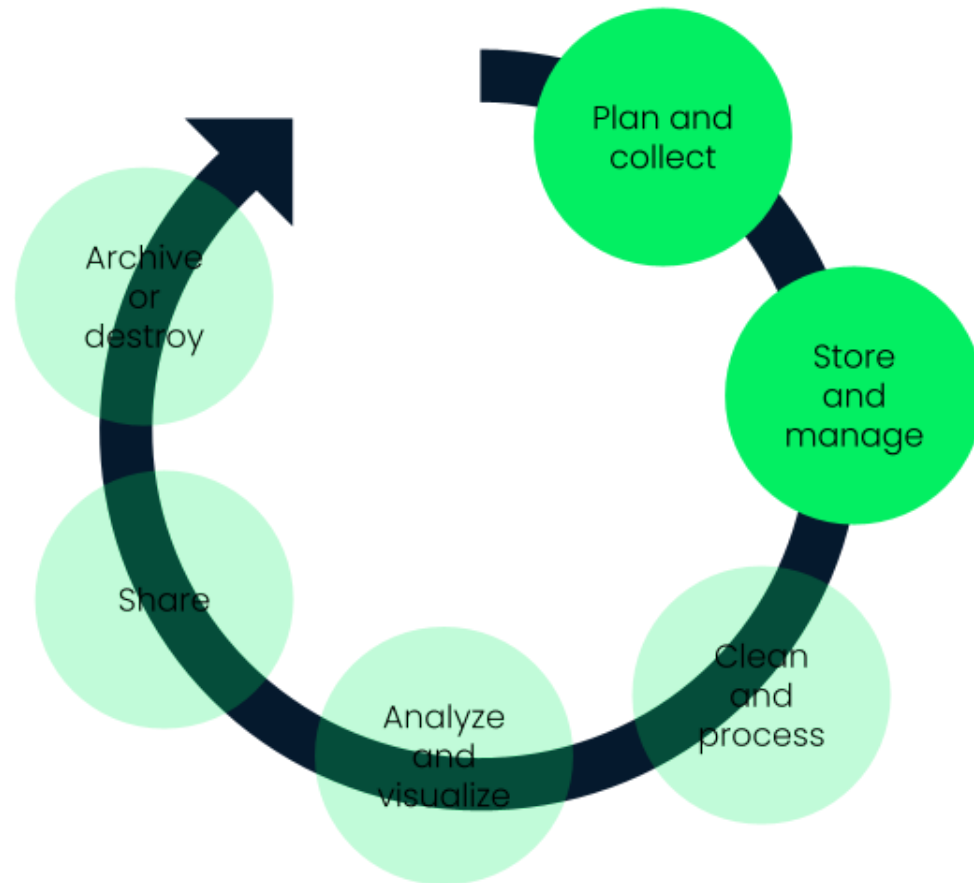


Plan and collect



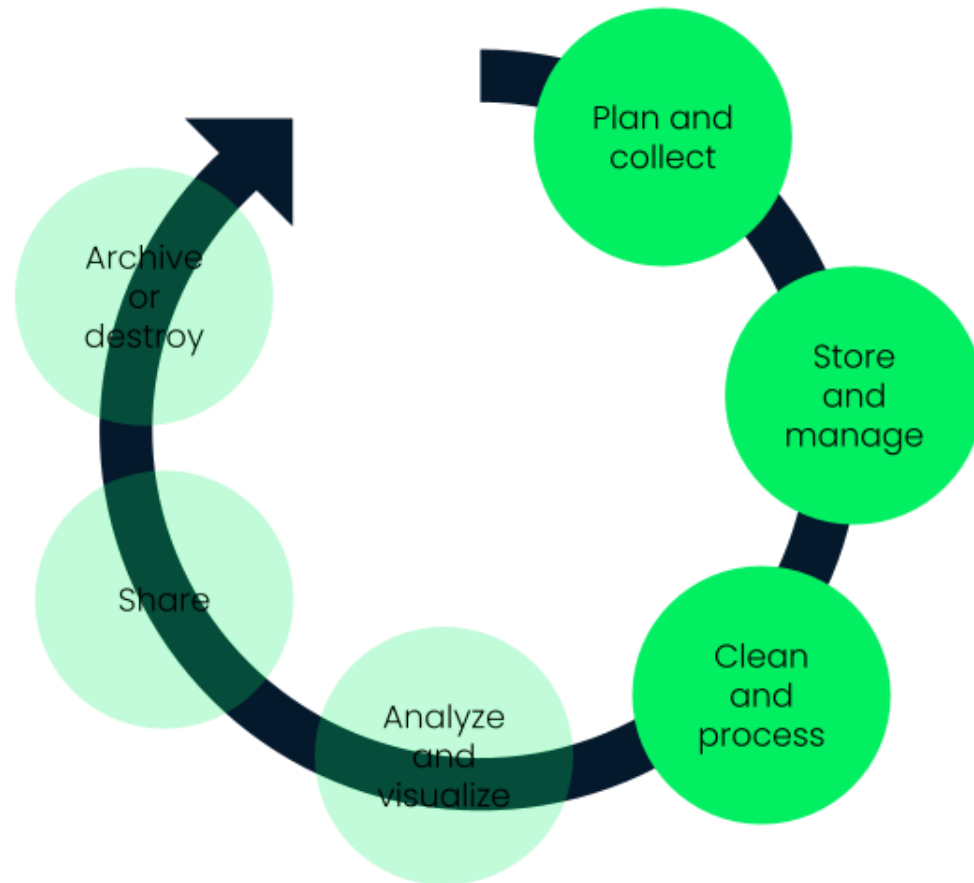
- Planning for other stages of the data life cycle
- Sharing of roles and responsibilities
- To achieve optimal results in terms of time and cost
- Collect or create data?

Store and manage



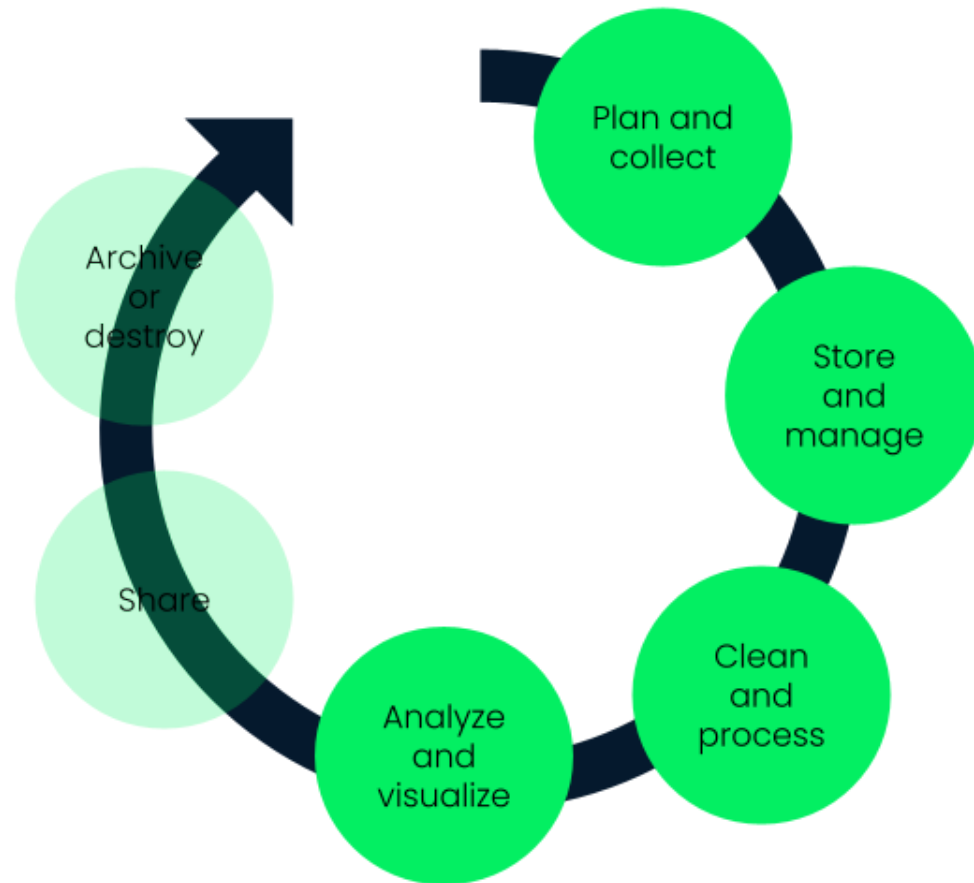
- Data stored in databases or data warehouses
- Easily accessible to the right person
- Removal of PII

Clean and process



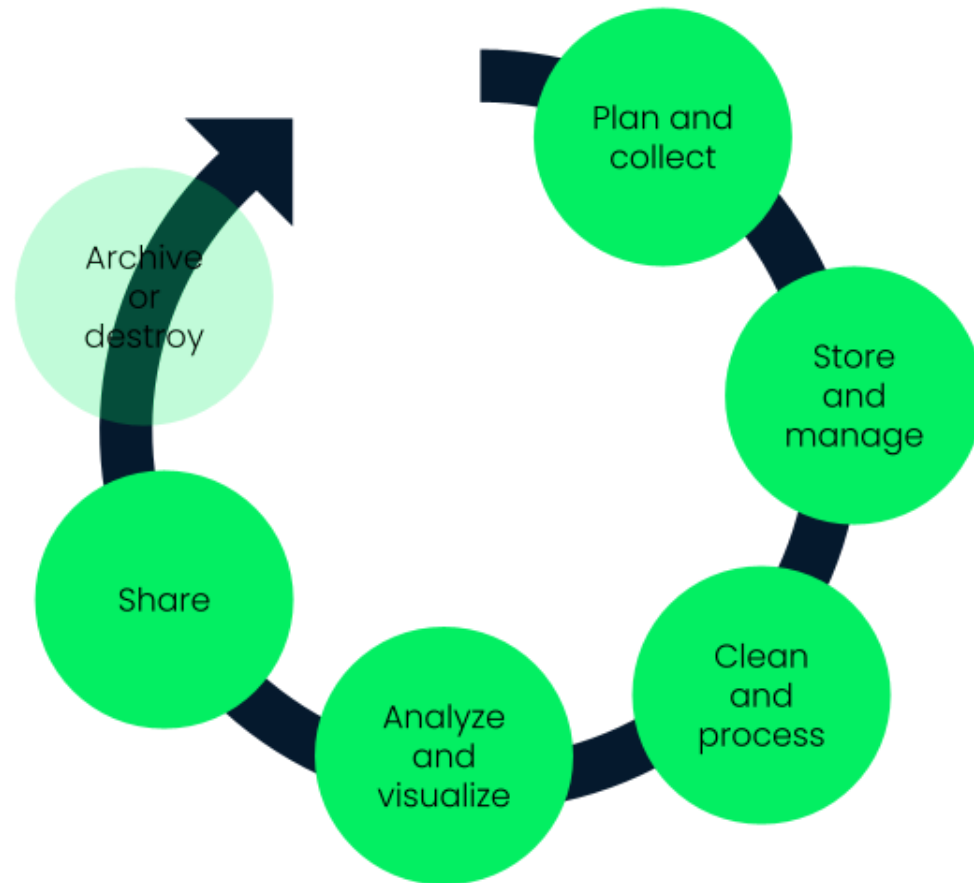
- Formatting data
- Missing data
- Transforming data

Analyze and visualize



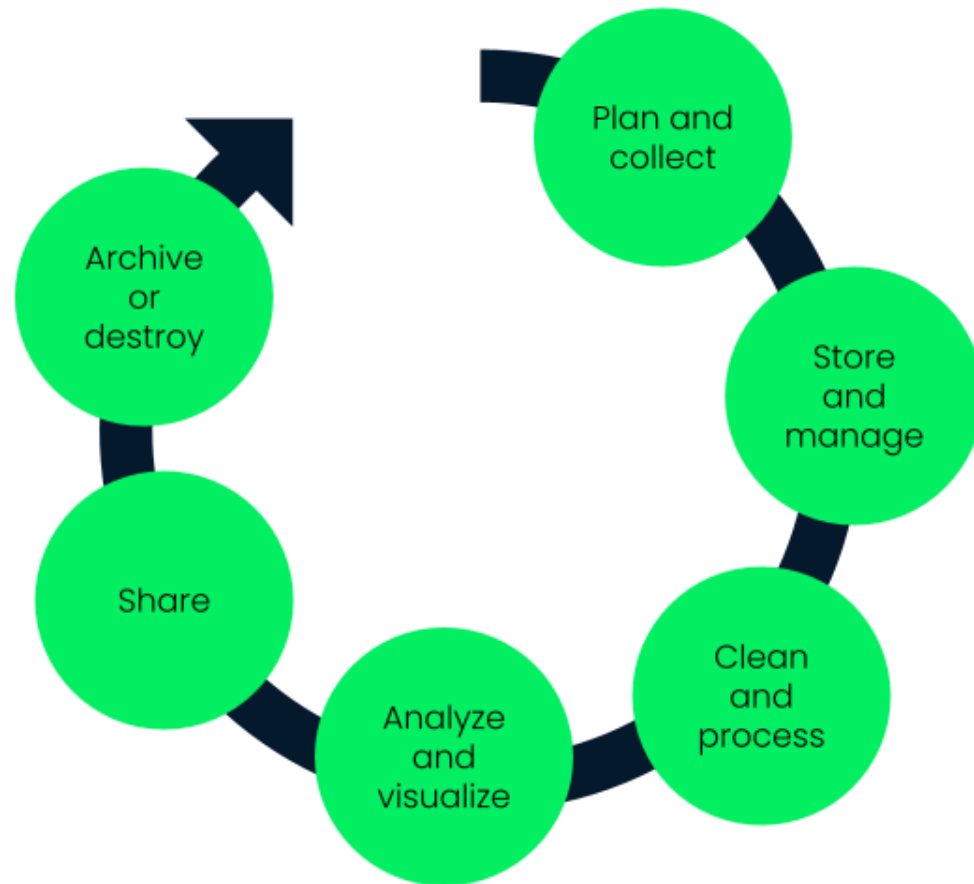
- Analyzing raw data for new insights
- Data is easier to interpret when visualized
- Analyzing and visualizing data is done in various ways

Share



- Communicate your results
- Dashboards, reports, papers

Archive or destroy



Decide if data should be kept or deleted

Data archiving:

- Data backups
- Documentation
- Digitizing

Data destruction:

- In rare cases
- Protecting private information
- Resources can be freed up

Let's practice!

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Common data mistakes

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Common mistakes about data

- Not having a clear goal or question
- Insufficient or wrong data
- Lack of appropriate analysis
- No clear communication of results

Carefully **plan** the data analysis process



Not clearly defining the problem

"Did you buy anything in the last month?"

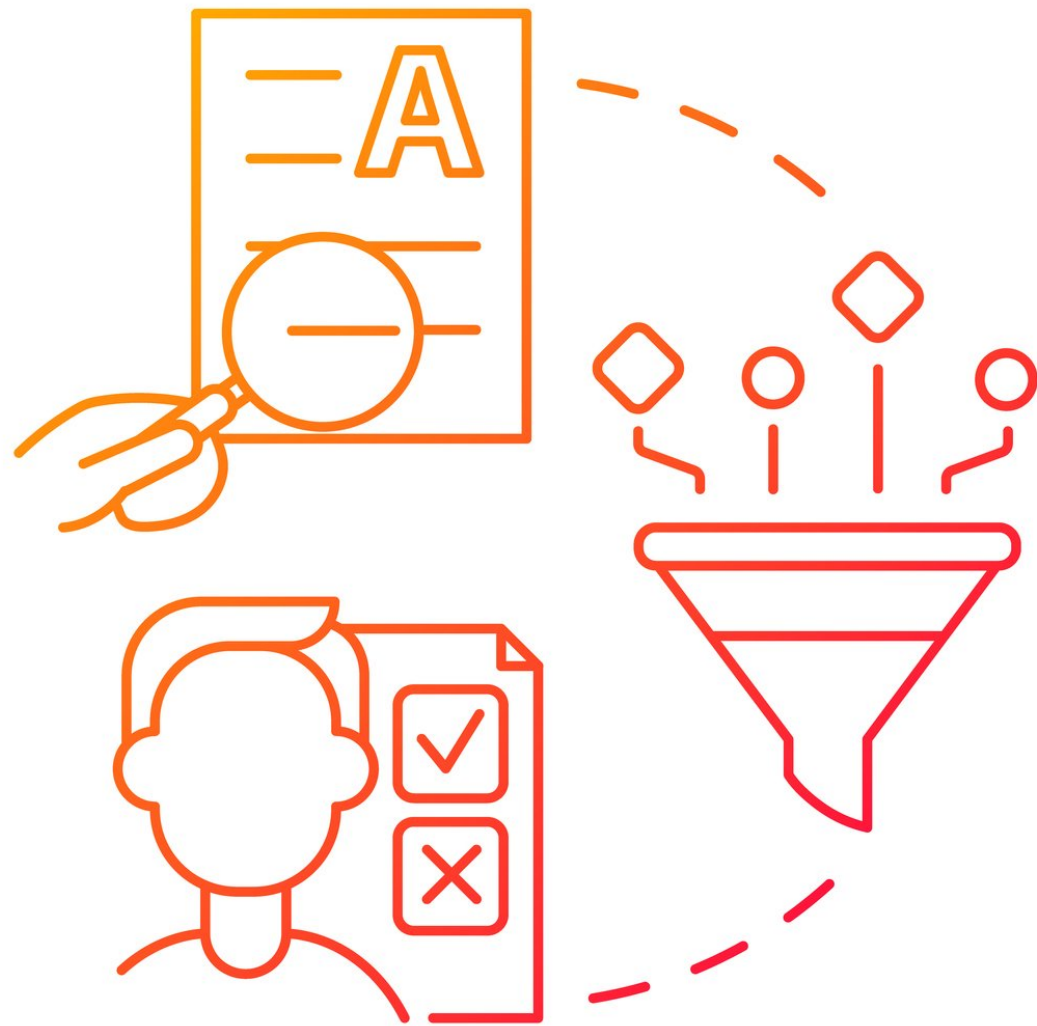
"Where did you make your last purchase?"

"Which payment method did you use?"

May lead to inappropriate data collection,
analysis, and conclusions



Insufficient or wrong data



Data bias: the data sample doesn't represent all the data

- Collecting the wrong data doesn't allow you to answer the research question
- Data still needs cleaning before analysis

Lack of appropriate analysis

- Jumping to conclusions too quickly
- **Lack of context:** a missing reason explaining the results
- Other examples include
 - Incorrect aggregations and calculations
 - Confusing correlation with causation



No clear communication of results



- Most **valuable** part of data life cycle
- Could lead to misunderstandings or incorrect conclusions
- Examples:
 - Too technical
 - Cherry-picking data points
 - Unclear visualizations

Let's practice!

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Wrap-up

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Chapter 1

- (Un)structured data
- Qualitative/quantitative data
- Data context

- Data is valuable in various fields
- Data-driven decision making
- Changes in data storage



Chapter 2

- DIKW framework
- Decision making process
- Data aggregations



Chapter 3

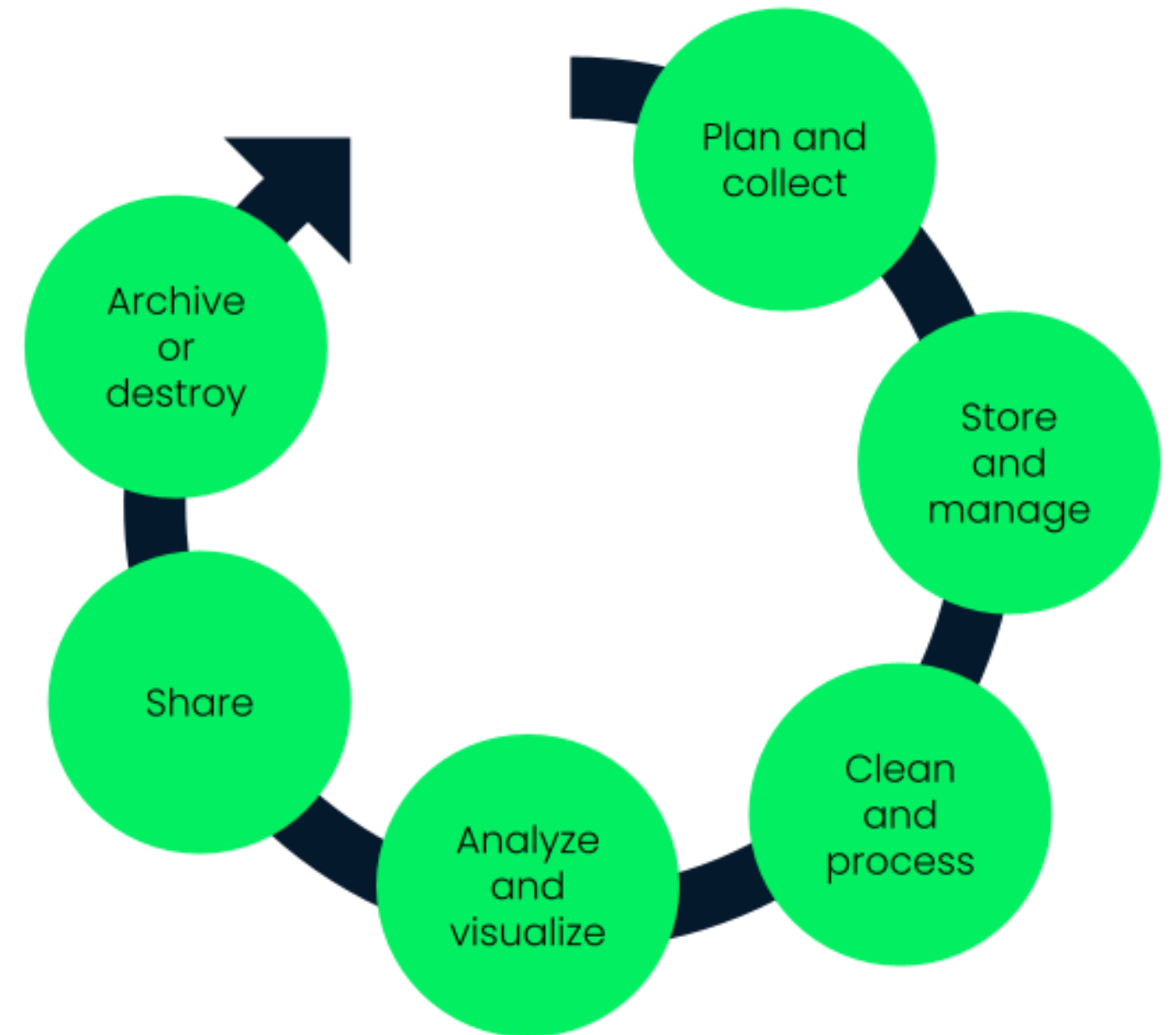
- Data ethics
- Data life cycle
- Common data mistakes

Understanding GDPR

Understanding Data Visualization

Introduction to Data Literacy

Communication Data Insights



Congratulations!

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